



Society of Petroleum Engineers

SPE-229689-MS

From 36 Hours to 15 Minutes: Revolutionary Reduction of Backreaming Time in 12.25" Hole Section in UAE Unconventional Wells

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Unconventional (UC) wells, given their narrow economic margins and short production lifespans, require strict cost control at every stage of the operation. Drilling, as one of the most significant contributors to well cost, becomes a key target for efficiency improvements. This paper addresses the recurring challenge of excessive back reaming time in the 12.25in hole section of UC wells in Southwest UAE—a problem that has, in some cases, resulted in stuck pipe incidents. The aim was to develop a consistent, field-proven approach to significantly reduce back reaming time and eliminate related operational risks.

The strategy combined planned field trials with data-driven evaluations. Leveraging the advantages of pad drilling, multiple wells were drilled under comparable geological conditions, with similar well profiles enabling a systematic variation of bottom hole assembly (BHA) configurations and mud systems. Trials included different combinations of roller, concentric and eccentric reamers, as well as mud types ranging from basic brines to engineered water-based fluids. Available offset well data, including mud logs and drilling parameters records, were reviewed to guide the mud properties and density definition.

Back reaming historically required up to 36 hours per well in this section — accounting for as much as 10% of total drilling time — making it a significant barrier to achieve the 15-day target for drilling 17,000 ft MD UC wells. Through iterative testing, a configuration using two roller reamers, paired with the mud system, reduced back reaming time to as little as 15 minutes. This approach not only eliminated stuck pipe incidents but also delivered substantial time and cost savings, improving the overall efficiency and economic viability of the project.

This case study represents a first-of-its-kind result in the UAE's unconventional operations, demonstrating that real-time field adaptation—through smart tool selection and mud optimization—can deliver step-change performance. The methodology is practical, repeatable, and applicable to other UC developments facing similar challenges.

Introduction

Unconventional (UC) wells present distinct challenges in both drilling and production, particularly due to their tighter economic margins and relatively short lifecycle. In the context of the UAE, where UC

developments are gaining traction, minimizing non-productive time (NPT) is critical to ensuring project viability. One recurrent bottleneck in these wells is the excessive time spent on back reaming, especially in the 12.25" intermediate hole section. Historical operations have shown that back reaming can consume up to 36 hours per well, contributing to nearly 10% of the total drilling time. In some cases, this also led to stuck pipe incidents, increasing operational risk and cost. This paper presents a field-validated approach that reduced back reaming time to as low as 15 minutes through a combination of BHA reconfiguration, mud system optimization, and pad drilling-based iteration.

Field Context and Operational Challenges

The wells presented in this study are located in Southwest UAE and are part of a broader unconventional development program targeting a multi-layered carbonate-dominated stratigraphy. The vertical 12.25" hole section penetrates through a complex interval composed of both mechanically competent and highly interbedded formations, each contributing to distinct drilling challenges.

Two primary fields, Field 1 and 2 as shown in Figure 1—share similar geological frameworks but exhibit variation in compressive strength, formation impact behavior, and structural layering. The lithostratigraphy across both fields includes six main formations, from M-1 near surface to F-1 at total section depth.

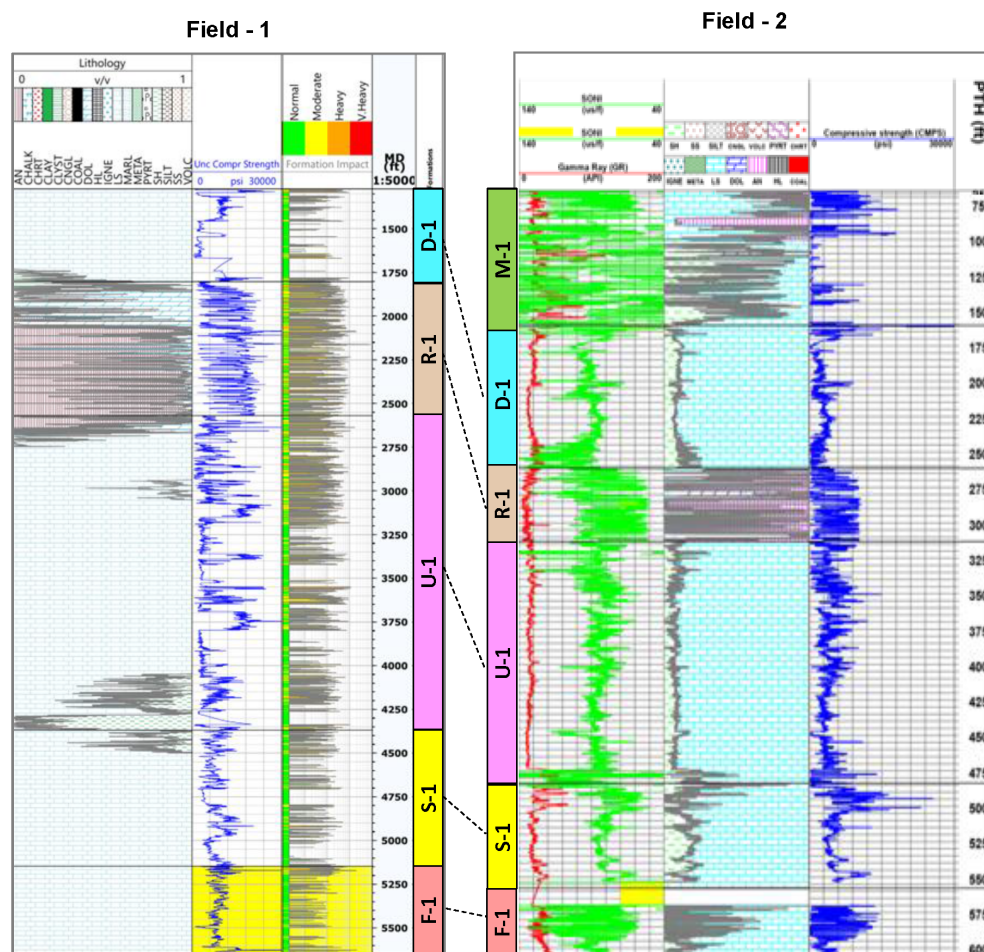


Figure 1—Comparative stratigraphy column for Field-1 and Field-2 in Southwest UAE

Formation Characterization

- D-1 (Calcareous Shales and Bioclastic Limestones): Characterized by shallow marine carbonates, this uppermost interval transitions from argillaceous limestones to bioclastic grainstones and

dolomites. It presents a moderate risk for dynamic losses during circulation due to variable permeability and mud interaction near the tidal-flat facies.

- R-1 (Thick Anhydrite Section): Composed predominantly of interbedded anhydrites and shallow-water carbonates, this unit is mechanically hard but highly layered, which increases the likelihood of shock and vibration, particularly with stiffer BHAs. This section is also where directional tool failure and bit bounce have historically been observed.
- U-1 (Thin Shale-Carbonate Sequence): The uppermost part of this zone contains bioclastic packstone-wackestones and argillaceous carbonates. The highly interbedded nature of this unit contributes to high shock and vibration response, particularly during bottom-up cleaning and back reaming. Additionally, dynamic mud losses have been intermittently observed due to poor hole cleaning and unpredictable breakouts.
- S-1 (Bioclastic Packstones with Coral Rudstone): A mixed energy depositional setting led to an accumulation of bioclastic packstones and wackestones, including coral fragments and dolomitic build-ups. The top of this section is also highly interbedded, further exacerbating vibrational responses. Tool fatigue and premature wear have been recurrent in this zone. Hole enlargement has been noted in prior caliper logs.
- F-1 (Basinal Marls and Marly Shales): This deep-water unit is marked by cemented shale lenses within argillaceous limestones, with indicators of calcareous microfauna (e.g., calcispheres, planktonic foraminifera). The unit's cemented intervals cause extremely slow rates of penetration (ROP) and have required either optimized WOB/BPM strategies or high-torque motors to maintain progress. In many offset wells, this interval also coincides with formation swelling and drag increases during tripping and back reaming.

Additionally, in Field-1, unconfined compressive strength (UCS) logs highlight sharp transitions from softer carbonates to high-strength zones exceeding 15,000 psi, which significantly impact vibration behavior during reaming and tripping out. In contrast, Field-2 displays more uniform Gamma Ray and UCS profiles, which allowed better control over tool response in successive wells.

These subsurface conditions created an operational environment that required to POOH with back reaming; the back reaming operations were associated with significant risks including tool fatigue, blade damage, and reduced BHA integrity where back reaming was not only time-consuming but also risk-prone, often resulting in excessive tripping time, BHA wear, stuck pipe events, and reaming-induced torque spikes. Specifically, the interaction between poorly tailored mud systems and vibrationally sensitive lithologies, such as interbedded anhydrites and laminated shales, contributed to low borehole quality and back reaming inefficiencies. The variability across lithological boundaries (e.g., R-1 to U-1) also required enhanced formation-specific planning, particularly in selecting the correct reamer type and mud properties.

A summarized version of this characterization is presented the [Table 1: Summary of formation risks](#).

Table 1—Summary of formation risks

TOPS	LITHOSTRATIGRAPHY	LITHOLOGY	HAZARDS
D-1	Calcareous shales and argillaceous limestones that grade upwards to peloidal, bioclastic pack stone-grain stone and dolomites of tidal-flat origin	Shallow marine carbonates Shale and argillaceous carbonates	Potential dynamic losses
R-1	Thick development of nummulitic shallow-water carbonates	Anhydrite	Highly interbed increase on shock and vibration
U-1	Thin basal bed of shales, capped by bioclastic pack stone-wackestones, argillaceous limestones and dolomitic limestones which were deposited in a shallow-marine setting.	Shallow marine carbonates Shale and argillaceous carbonates	Top section is highly interbedded — increases shock and vibration Potential dynamic losses CHERT – non PDC drillable
S-1	Peloidal, bioclastic packstones and wackestones and dolomitic limestones with coral and rudist build-ups which accumulated on a mixed, shallow, sub-tidal to tidal-flat setting	Shallow marine carbonates	Top section is highly interbedded — increases shock and vibration Potential dynamic losses Formation swells
F-1	Deep-water basinal marls, argillaceous limestones and marly shales with a deep-marine fauna (calcspheres and planktonic foraminifera)	Shale and argillaceous carbonates	Cemented shale in limestone – slow ROP

Given the complex and variable lithological profile described above—particularly the interbedded nature of the U-1 and S-1 formations and the mechanical challenges posed by the F-1 interval—it became clear that a single, static drilling approach was insufficient to mitigate the risks encountered in the 12.25" hole section. Despite minor tool and fluid adjustments in earlier wells, operational inefficiencies persisted, including extended back reaming times and increased vibration. To address these issues, a more rigorous, structured, and data-guided optimization strategy was required. The following section outlines the methodology used to design and implement field trials across pad-drilled wells, focusing on tool configuration, fluid properties, and formation-specific responses—ultimately enabling real-time adaptation and performance improvement.

Methodology and Optimization Strategy

Downhole sensor deployment

To capture real-time downhole dynamics and evaluate drilling performance across various formations, a high-frequency memory-mode sensor [Figure 2](#) was deployed directly above the bit. The tool used was a miniature autonomous module with internal power supply and onboard storage, capable of recording multiple drilling mechanics parameters without requiring surface telemetry.

The sensor measured 3-axis shock (± 200 g) and vibration (± 16 g) with a sampling rate of 800 Hz, while also capturing rotational speed (± 666 rpm) and torsional acceleration (up to 6.4×10^6 deg/s²) at 1,125 Hz. These high-resolution signals allowed identification of transient dysfunctions—particularly axial and lateral shocks transmitted through the bit and the lower BHA. Additionally, temperature logging (from -40 to 125 °C) and total recording time of up to 200 hours ensured full coverage of the 12.25in section.



Figure 2—Downhole sensor

The data was processed through automated workflows post-run, enabling clear visualization of the interaction between BHA components and lithological transitions. This formed the basis for evaluating tool performance, identifying zones with excessive mechanical stress, and correlating downhole behavior with surface drilling parameters.

BHA Configurations

A series of BHA configurations were trialed across multiple wells in the 12.25" section to assess their impact on backreaming performance. Each design aims to balance hole conditioning, tool durability, and mechanical stability in a vertical, interbedded carbonate-shale environment. All assemblies incorporated standard directional components—PDC bit, rotary steerable system (RSS), and MWD tools—followed by different reaming and stabilization strategies.

Three main BHA configurations were tested as shown in [Figure 3](#).

- BHA #1: Roller Reamer Configuration

This configuration included two 12.25in mechanical roller reamers, spaced apart in the drill string to improve borehole conditioning over a longer interval. The roller reamers provided mechanical stabilization and damping, particularly beneficial in interbedded or shock-prone zones. Float subs, NMDCs, and HWDP were used for stiffness.

- BHA #2: Eccentric Reamer Configuration

This assembly used a 12.25in eccentric reamer with PDC bit designed to improve trajectory stability and hole condition. Dynamic stabilization was expected, but the tool showed limited benefit in highly variable lithologies. The eccentric blade profile required close monitoring for wear, especially in abrasive formations.

- BHA #3: Concentric Reamer Configuration

This BHA used a 12.25" concentric reamer with fixed spiral blades and undergauge design. Positioned close to the bit, it targeted borehole cleaning and enlargement near the well bottom. Though mechanically robust, it lacked the same damping effect as the roller-based design.

BHA #1 Roller Reamer				BHA #2 Eccentric Reamer				BHA #3 Concentric Reamer			
#	Description	Max OD (in)	Length (ft)	#	Description	Max OD (in)	Length (ft)	#	Description	Max OD (in)	Length (ft)
1	12 1/4" PDC Bit	12.250	1.08	1	12 1/4" PDC Bit	12.250	1.00	1	12 1/4" PDC Bit	12.250	1.00
2	RSS 12 1/4" w/ Stab	12.125	13.90	2	RSS 12 1/4" w/ Stab	12.125	12.34	2	RSS 12 1/4" w/ Stab	12.125	14.00
3	MWD Rec	7.930	3.25	3	Short Hop	8.250	7.10	3	MWD Rec	8.000	3.25
4	MWD	8.000	30.94	4	MWD	8.410	27.15	4	MWD	8.000	30.27
5	8" NMDC	8.000	30.95	5	8 1/4" NMDC	8.250	30.59	5	8" NMDC	8.000	29.35
6	8.25" Float Sub	8.250	3.94	6	8.25" Filter Sub	8.000	5.54	6	8.25" Float sub	8.250	2.40
7	8.25" Filter Sub	8.250	5.59	7	12.125in String Stab	12.125	5.46	7	8.25" float valve	8.250	2.00
8	12 1/4" Roll Reamer	12.250	5.46	8	Float sub	8.250	1.75	8	8.25" Filter Sub	8.250	5.27
9	1 x 8" DC	8.000	30.61	9	8 1/4" DC	8.250	31.00	9	12.125in String Stab	12.125	6.89
10	12 1/4" Roll Reamer	12.250	7.08	10	12.25 Ecc Reamer	12.250	4.90	10	8 1/4" DC	8.250	30.85
11	9 x 8" DC	8.000	276.62	11	9 x 8 1/4" DC	8.250	281.40	11	12-1/8" Conc Reamer	12.125	6.70
12	Drilling Jar	8.250	23.32	12	Drilling Jar	8.500	29.10	12	9 x 8" DC	8.250	276.21
13	3 x 8" DC	8.000	92.32	13	3 x 8 1/4" DC	8.250	94.07	13	Drilling Jar	8.000	22.53
14	Crossover	8.000	3.61	14	Crossover	8.250	4.05	14	3 x 8" DC	8.250	92.18
15	15 x 5 1/2" HWDP	7.000	465.71	15	12 x 5 1/2" HWDP	7.000	369.25	15	Crossover	8.250	3.60
16	5-1/2" 21.90 DP	6.625	30.00	16	5-1/2" 21.90 DP	6.625	30.00	16	15 x 5 1/2" HWDP	7.000	467.00
								17	5-1/2" 21.90 DP	6.625	30.00

Figure 3—BHA configurations

In addition to mechanical optimization, a high-frequency downhole dynamics Figure 4 recording tool was deployed in several wells to better understand how vibrational energy was transferred through the BHA. The downhole sensor was positioned in the bit, allowing for real-time recording of drilling dynamics inside the bit. This helped distinguish between formation-induced shock and string-induced movement.

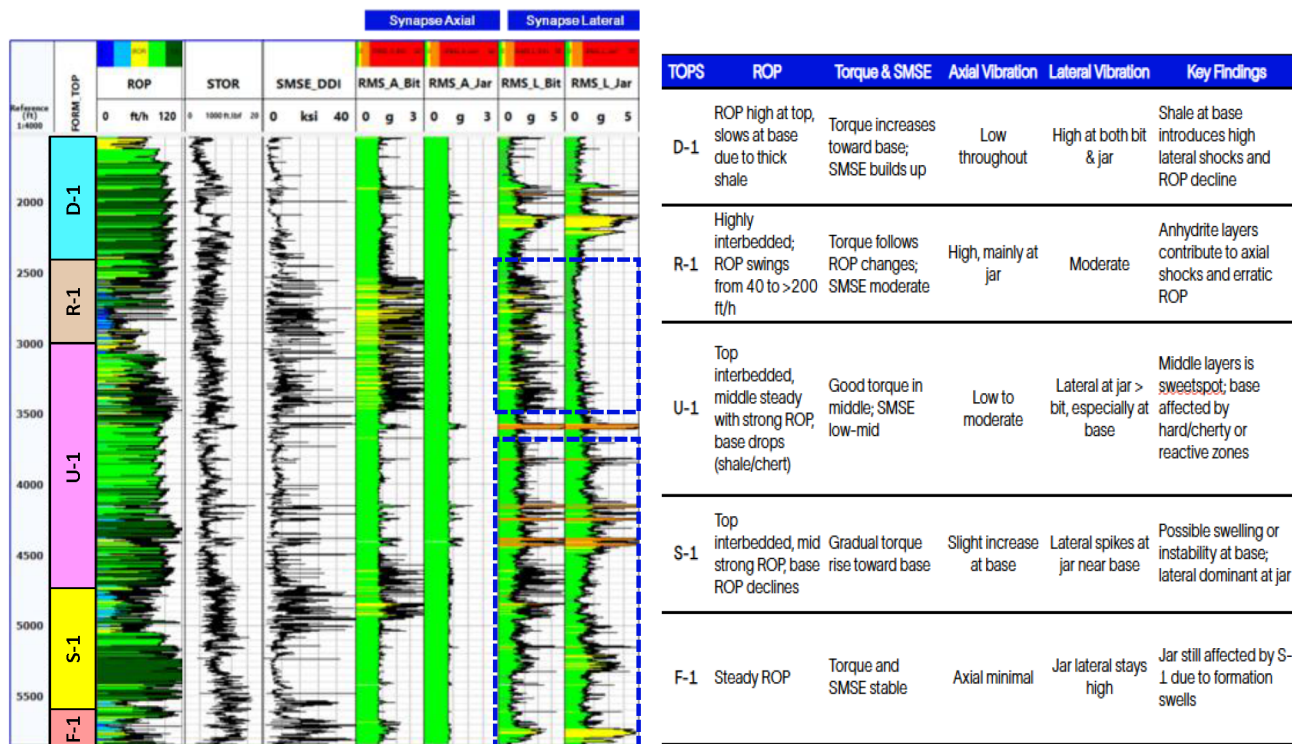


Figure 4—High-frequency downhole dynamics log

The data revealed that axial vibration was generally low, except in highly interbedded zones like R-1 and U-1, where moderate axial shocks were observed, especially at the jar. Lateral vibrations were more prominent, particularly in the S-1 and D-1 intervals, often correlating with torque buildup and erratic ROP.

These observations confirmed that certain formations—like the base of D-1 and the top of S-1—trigger high lateral shocks, which likely contribute to borehole enlargement, tight spots, and the need for aggressive backreaming. This real-time insight reinforced the selection of roller reamers, which proved more effective at absorbing these vibrations and improving borehole conformity in affected zones.

Reamers Selection and field Evaluation

As part of the effort to improve borehole shape and reduce back reaming time, three different reamer types Figure 5 were tested throughout the campaign: roller, eccentric, and concentric. Each tool offered different mechanics for reaming and stabilization. The selection process aimed to identify the most effective tool to minimize tight spots, ledging, and drag while tripping out of hole.



Figure 5—Tested reamers

Roller Reamer. The roller reamer was trialed in multiple wells and ultimately delivered the most consistent results. Its full-gauge rollers provided continuous mechanical contact with the borehole wall, acting as both a stabilizer and a smoothing tool—particularly effective in interbedded formations like the U-1 and S-1 formations. It helped eliminate irregularities created during drilling and prevented cuttings beds from forming during connections or trips. In wells where two roller reamers were spaced apart in the BHA, the trip-out torque profile was notably smoother, and back reaming was minimum or no longer required in several cases.

Eccentric Reamer. The eccentric reamer was tested in a limited number of wells. It uses fixed blades with an offset rotation to gradually contact and condition the borehole wall. While the tool functioned as intended in softer formations, its behavior in the F-1 formation and lower sections was less predictable. In particular, the dynamic movement of the blades appeared to cause side-cutting and localized enlargement in tight transitions, leading to high localized drag plus create additional vibrations in the BHA. Wear was also observed on the blades after limited footage and in some cases damaged the directional tools. This setup required back reaming in every case and did not lead to noticeable improvements in borehole condition.

Concentric Reamer. While mechanically robust, the tool did not contribute to improved hole shape over extended intervals. Its impact was limited to localized conditioning near the bit, and it lacked the damping or stabilizing effect seen with the roller design. In interbedded zones, this setup still required full back reaming, and several wells experienced high trip-out torque even after unplanned intermediate trips.

Among the three tools evaluated, the roller reamer consistently delivered the best results. Its ability to mechanically smooth the borehole and provide effective stabilization across longer intervals directly contributed to the reduction—or complete elimination—of back reaming in several wells. The eccentric and concentric tools, while functional in certain conditions, failed to address the wellbore irregularities that typically form in the interbedded zones of the 12.25in section. As a result, they offered limited value in improving trip-out performance compared to the roller-based configuration.

Mud Systems

Three different mud strategies were applied across the campaign to support the 12.25" vertical hole section, each adjusted based on observed performance during drilling and back reaming. While mud formulation was not the primary focus of this study, its influence on hole condition, clay inhibition, and cuttings suspension was critical in supporting the mechanical strategy that ultimately reduced backreaming time.

Formulation 1 – Low Rheology, Light Weight, Relaxed Fluid Loss. The first mud recipe was selected as a conservative starting point, based on offset performance and formation behavior in the upper part of the 12.25" section. It used a KCl polymer WBM with 3% KCl, mud weight of 9.0–9.2 ppg, and low rheology (YP 10–15 lbf/100ft²). API fluid loss was kept relaxed (<25 cc/30 min) to simplify system maintenance.

This approach aimed to minimize losses in the S-1 formation, known for natural fractures, while also accounting for limited water resources & availability. However, high ROPs made it difficult to build mud properties fast enough, and the light, relaxed system led to poor borehole conformity in interbedded zones. As a result, backreaming was still prolonged and inefficient, leading to a shift toward stronger mud properties in later wells.

Formulation 2 – Higher Rheology, Tight Fluid Loss, Same Density. In this approach, the KCl concentration (3%) and mud weight (9.0–9.2 ppg) remained unchanged, but rheology was increased (YP 22–28 lbf/100ft²) and API fluid loss was reduced to <6 cc/30 min. The goal was to improve cuttings suspension and reduce filtrate invasion in the deeper, more reactive formations.

While some improvements in torque consistency were observed, this setup still failed to deliver clean borehole conditions during trips. Backreaming remained long, as the system lacked the inhibition and density needed to fully stabilize the well in interbedded and swelling zones.

Formulation 3 – Higher Rheology, Increased Inhibition, Moderate Fluid Loss. This final configuration introduced 8% KCl for enhanced inhibition, YP of 22–28 lbf/100ft², and a slightly higher mud weight of 9.4 ppg. Fluid loss was kept moderate (<18 cc/30 min) from the beginning of the section to top of F-1 and tighter Fluid loss < 6 cc/30 min from F-1 to TD.

Results and Comparative Analysis

Case Study

Eccentric Reamer Evaluation. In the first field trial [Figure 6](#), Well A and B, two 12¼" sections were drilled using the same BT616 PDC bit but with two different types of eccentric reamers (Type #1 and Type #2). Both reamers were able to drill the entire 12.25in section with acceptable borehole quality; however, the performance statistics highlighted a clear compromise between stability and efficiency.

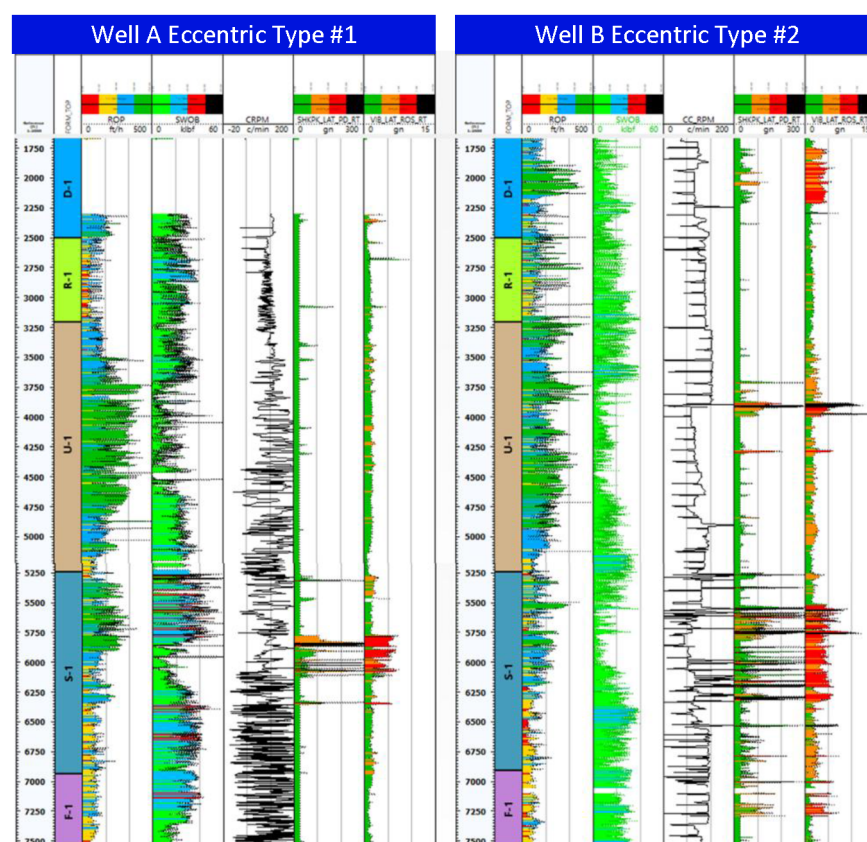


Figure 6—Eccentric BHA Performance

Eccentric Type #1, although providing more controlled drilling dynamics with lower lateral and axial vibration, required significant backreaming and resulted in longer section time, indicating conservative hole opening behavior.

Eccentric Type #2, while aiming for a more aggressive approach and eliminating backreaming, still showed higher shock levels, especially in the S-1 and F-1 formations, along with severe stick-slip—a sign of increased stress at the reamer and bit interface.

Despite the reduced overall section time observed with Type #2, both eccentric reamers failed to deliver optimal ROP performance, which was notably lower than expected. The ROP degradation, combined with higher mechanical energy input, suggests that while borehole stability was improved, it came at the cost of drilling performance.

This finding highlights a key limitation of eccentric reamer technology: although it helps mitigate downhole vibrations and assists in hole conditioning, it does not translate into efficient drilling performance. This sets the stage for evaluating roller-based reamer technologies in the next trial.

Roller Reamers. In this comparative study Figure 7, presenting four batch-drilled wells on a drilling pad with 10 meters spacing. Well #1, drilled with an eccentric reamer, showed clear signs of instability—marked by sharp torque spikes and consistently elevated SMSE values, especially through formation transitions like U-1 to S-1. These responses pointed to poor energy transfer and increased downhole vibrations, likely caused by the uneven borehole geometry introduced by the eccentric reamer design.

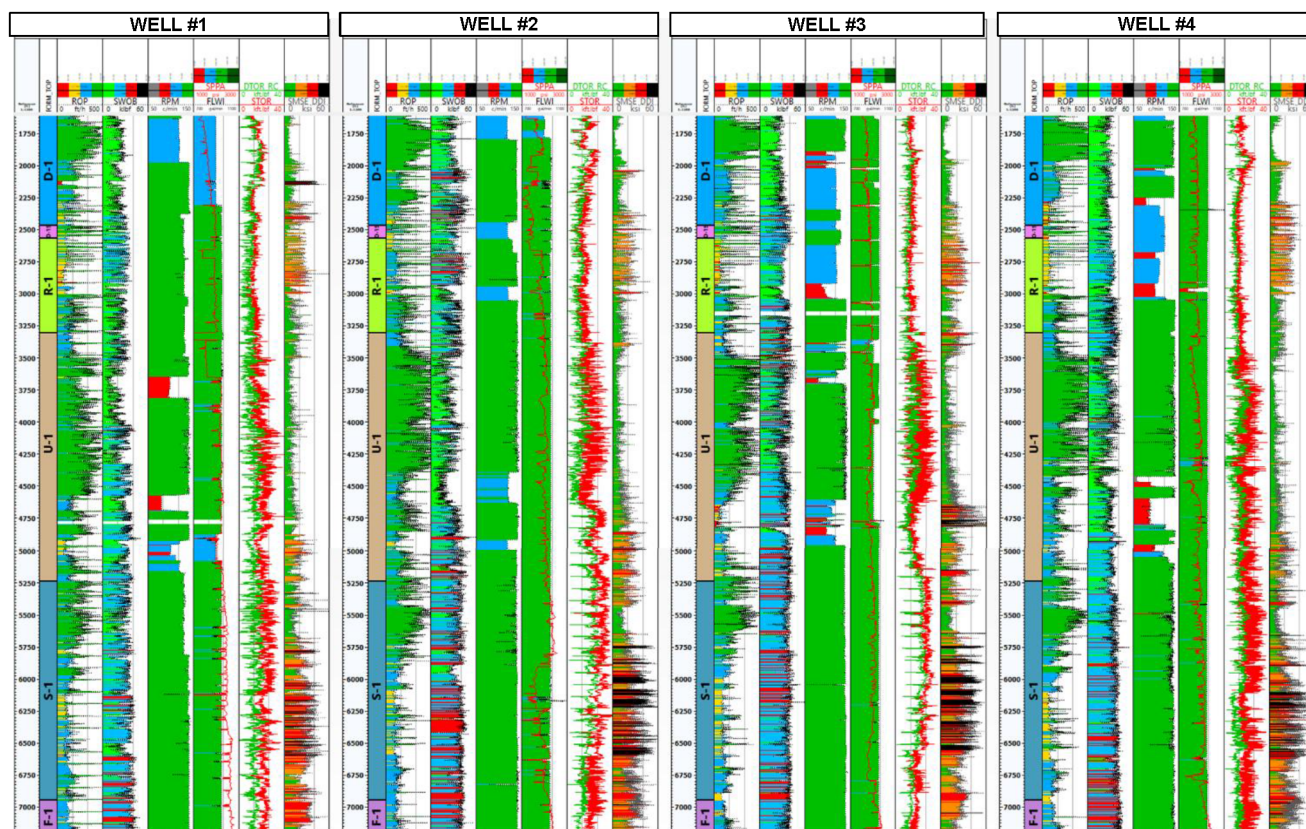


Figure 7—Drilling Dynamics Pad 4

By contrast, Wells #2, #3, and #4, which were all drilled with roller reamers using the same BHA configuration, demonstrated a much more stable drilling performance. The torque response was smoother, SMSE trends were cleaner, and the ROP remained steady across the same formations. These wells required minimal backreaming—less than an hour—compared to 36 hours spent on Well #1, further confirming better hole condition and improved string behavior.

This contrast is even more evident when observing trip-out (POOH) torque behavior across the same stratigraphic intervals. In Well #1, drilled with the eccentric reamer, the transition from S-1 into U-1 required 36 hours of aggressive backreaming, with clear signs of erratic torque and unstable SRPM, pointing to poor hole quality and possible spiraling. In comparison, Wells #2, #3, and #4—each drilled with roller reamers—navigated the same sections with less than one hour of backreaming and exhibited smooth torque profiles with minimal drag. These trends highlight not only improved hole cleaning and geometry with roller reamers, but also a more efficient and controlled string movement—further validating the dynamic benefits of a stable borehole.

Overall, the results clearly suggest that roller reamers provided better dynamic stability and hole quality, allowing the team to drill more efficiently and with fewer issues. While the eccentric reamer may have delivered some hole-enlargement benefits, it came at the cost of drilling performance and tool reliability—outweighed by the consistent results achieved with roller reamers.

Discussion

The comparative performance [Figure 8](#), [Figure 9](#) across the six wells demonstrates a consistent correlation between reamer type and overall downhole behavior. Wells A and B, both drilled with eccentric reamers under controlled parameters, achieved trip-out speeds of up to 9 stands per hour. However, despite these

controlled conditions, the on-bottom ROP remained limited ~ 100 ft/hr, highlighting a trade-off between borehole conformity and drilling performance.

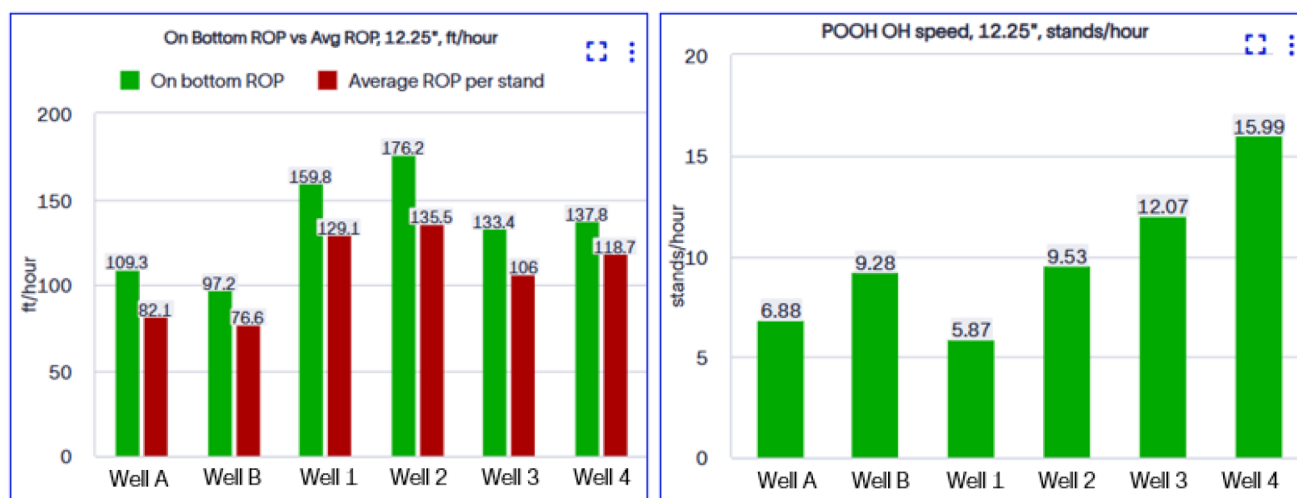


Figure 8—Pad 4 Performance – ROP, POOH speed

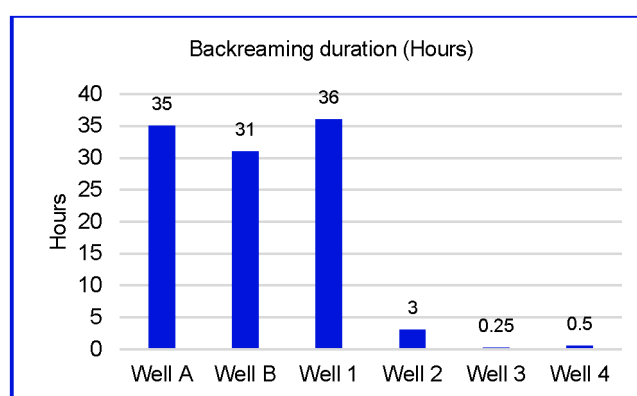


Figure 9—Backreaming duration while POOH in 12-1/4" hole section

Well #1, also drilled with an eccentric reamer, was executed with a performance-driven strategy aiming to increase efficiency. Although this approach succeeded in maintaining higher ROPs—peaking above 160 ft/hr—it came at the expense of borehole quality. This was evidenced by erratic torque spikes, elevated SMSE trends, and a prolonged backreaming duration exceeding 36 hours. Such indicators point to poor energy transfer and string instability, likely caused by reamer-induced spiraling or inconsistent borehole gauge, which led to inefficient cuttings evacuation and impaired string movement.

From Well #2 onward, the use of roller reamers combined with optimized fluid loss control resulted in significant improvements. Initial runs employed a tight fluid loss strategy throughout the entire interval (API FL < 6 cc), yielding better hole cleaning and improved tripping speeds, although the full expected POOH performance was not yet realized. The approach was then refined: fluid loss was slightly relaxed from surface to the top of the F-1 formation, while maintaining tight control (API FL < 6 cc and 8% KLC) across the F-1 interval. This strategy enabled cleaner hole conditions, with zero backreaming and on-bottom ROP exceeding 130 ft/hr. In Well #4, this methodology was replicated successfully, achieving similar ROP and further improving POOH speed to 16 stands per hour.

The performance observed in Wells #2, #3, and #4—drilled using roller reamers under equivalent BHA and formation conditions—demonstrated smoother torque and SMSE profiles, negligible backreaming, and stable string dynamics, particularly across critical intervals such as the U-1 to S-1 transition. These outcomes

reinforce the mechanical advantages of roller reamers in enhancing borehole geometry, reducing downhole dysfunctions, and enabling more efficient and reliable drilling operations through interbedded and reactive formations.

Conclusions

The field data strongly supports the operational benefit of using roller reamers over eccentric reamers in the 12 ¼" section across the evaluated wells. Key takeaways include:

- **Enhanced Borehole Stability:** Roller reamers deliver smoother torque and SMSE trends, indicating better hole geometry and reduced dynamic dysfunction.
- **Reduced Backreaming Time:** Backreaming dropped from 36 hours in the eccentric reamer well to less than one hour with roller reamers, minimizing rig time and associated costs.
- **Improved String Mobility:** The reduced drag and consistent POOH conditions suggest that roller reamers contributed to improved axial and rotational movement of the BHA.
- **Consistent ROP across Transitions:** While eccentric tools introduced instability—requiring trade-offs in ROP—the roller reamer BHAs sustained steady penetration rates with fewer dysfunctions.
- **Fluid strategy:** API FL < 18 cc from the beginning of the section till top of F-1 formation followed by tight FL < 6cc in F-1 to section TD

The implementation of roller reamers in similar formations and operational setups is therefore recommended as a best practice to optimize drilling performance, mitigate vibration-related risks, and reduce overall non-productive time.

Acknowledgments (Optional)

The authors would like to thank the operating company, service providers, and their joint project teams, for their support and collaboration throughout this study, and for granting permission to publish these results. Special appreciation is extended to the drilling teams for their field expertise and operational insights, which were instrumental in the successful execution of this campaign. We also acknowledge the contributions of the real-time monitoring and engineering teams, whose diligent analysis of downhole dynamics significantly enriched the findings presented in this paper.

Nomenclature

Term / Abbreviation	Description
BHA	Bottom Hole Assembly
WBM	Water-Based Mud
KCl	Potassium Chloride
MW	Mud Weight (ppg)
YP	Yield Point (lbf/100ft ²)
API FL	API Filtrate Loss (cc/30 min)
SMSE	Specific Mechanical Specific Energy
ROP	Rate of Penetration (ft/hr)
POOH	Pull Out of Hole
UC	Unconventional
RR	Roller Reamer
ER	Eccentric Reamer

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